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MAC 283
HW#6

1. Suppose a computer using direct mapped cache has 2^{20} bytes of byte-addressable main memory, and a cache of 32 blocks, where each cache block contains 16 bytes.

- How many blocks of main memory are there?

Main memory/ cache size = blocks in main memory

$$2^{20} / 2^4 = 2^{16}$$

- What is the format of a memory address as seen by the cache, i.e., what are the sizes of the tag, block, and offset fields?

Tag = 11 bits Block = 5 bits Offset = 4 bits

- To which cache block will the memory address 0x0DB63 map?

Block 26

2. Suppose a computer using a fully associative cache has 2^{24} bytes of byte-addressable main memory and a cache of 128 blocks, where each block contains 64 bytes.

- How many blocks of main memory are there?

Main memory/ cache size = blocks in main memory

$$2^{24} / 2^7 = 2^{17}$$

- What is the **format of a memory address** as seen by the cache, i.e., what are the sizes of the tag, and offset fields?

Offsets = 7 bits Tag = 17 bits

- To which cache block will the memory address 0x01D872 map?

Block 8

21. Suppose we have 2^{10} **bytes of virtual memory** and **28 bytes of physical main memory**. Suppose the **page size is 24 bytes**.

- How many pages are there in virtual memory?

$$2^{10} / 2^4 = 2^6$$

- How many page frames are there in main memory?

$$2^8 / 2^4 = 2^4$$

- How many entries are in the page table for a process that uses all of virtual memory?

The number of entries for a process to use all virtual memory is the same as the amount of pages in virtual memory

20. Suppose you have a byte-addressable virtual address memory system with 8 virtual pages of 64 bytes each, and 4 page frames.

Assuming the following page table, answer the questions below:



- How many bits are in a virtual address?

$$2^3 * 2^6 = 2^9$$

- How many bits are in a physical address?

$$2^2 * 2^6 = 2^8$$

- What physical address corresponds to the following virtual addresses (if the address causes a page fault, simply indicate this is the case)?
 - 0x00

Physical Address: 0x00

- 0x44

Physical Address : 0x04

- 0xC2

Page Fault

- 0x80

Physical Address: 0xC0